HEAT TRANSFER PROJECT

Design of an Affordable
Liquid Cooling System for High-Power CPUs

ADVANCED THERMAL SOLUTIONS:

NEXT-GEN CPU LIQUID COOLING.

Group -5

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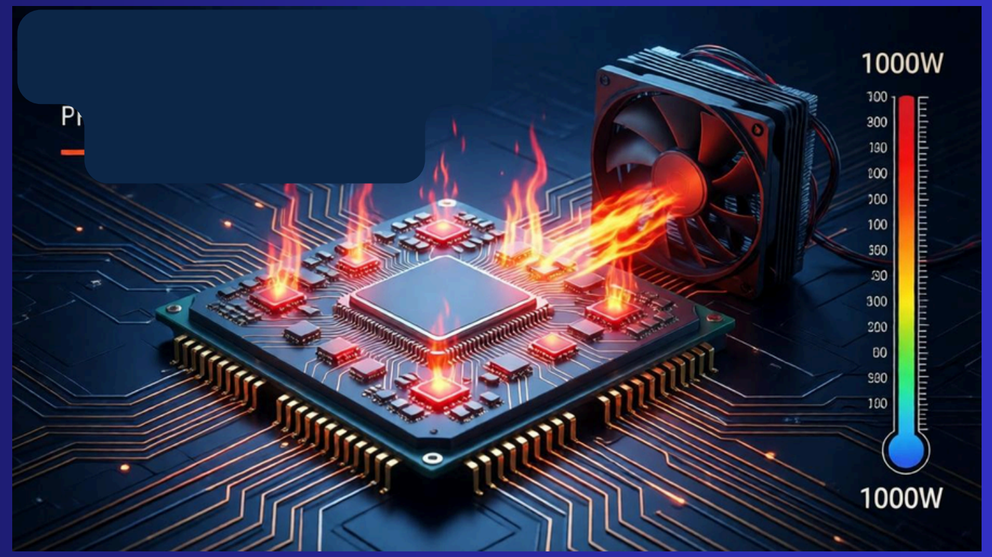
Problem and Problem Statement

Problem

- Modern CPUs reaching 100-1400W peak power
- Increasing power density → hotspot formation
- Air cooling ineffective beyond ~100–200 W/cm²
- Cooling consumes significant system energy



Problem Statement

Current cooling solutions either lack efficiency (air cooling) or are expensive and complex (advanced liquid systems). A scalable, efficient, and affordable solution is required.

Need Analysis:

Increasing Thermal Demand

- CPU TDP rising steadily over the past decade
- High-power processors generate localized hotspots

Performance & Reliability Impact

- Hotspots → Thermal throttling
- Excess temperature → Reduced lifespan
- Poor cooling → Increased operational cost

Industry & Innovation Trend

- Rising patent activity in microchannel and immersion cooling
- Growing market for liquid cooling systems
- Expansion of liquid cooling in gaming and HPC systems

Identified Gap

Existing solutions:

Air cooling → Limited heat flux handling

Premium liquid systems → Expensive and complex

publication

CPU Power & Heat Generation Statistics

Processor	TDP
Intel Core i9-14900K	125 W (Base), ~253 W (Turbo)
AMD Ryzen 9 7950X	170 W (Base), ~230 W (Boost)
AMD EPYC 7763 (Server)	280 W
NVIDIA H100 GPU (AI/HPC)	700 W

Heat Flux Density Data

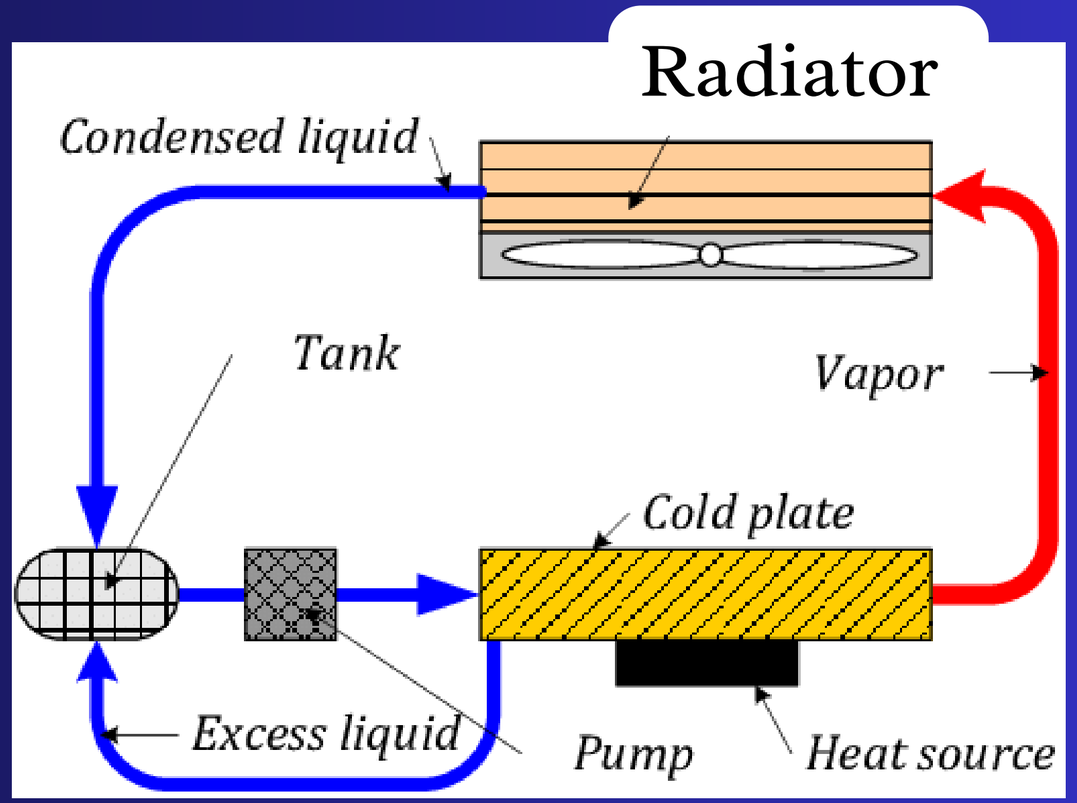
Technology	Heat Flux
Conventional air cooling limit	~100 W/cm²
Advanced liquid cooling	200–500 W/cm²
Two-phase microchannel cooling	>1000 W/cm²

patent

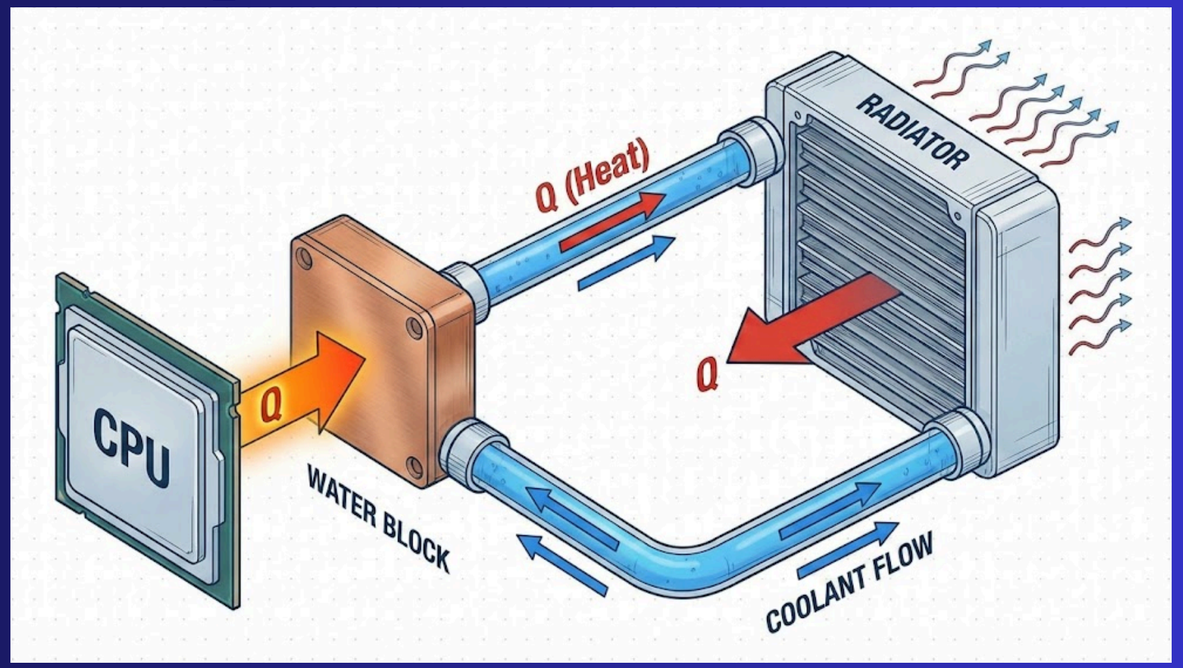
Solution Overview
Liquid Cooling System

Main Components:

1. Water Block (Copper Base, k=400 W/mK)
    a. Mounted on CPU
    b. High thermal conductivity
    c. Micro channel design for better heat transfer
2. Coolant
    a. Distilled Water(70%) + Ethylene Glycol(30%) mixture
    b. Non-conductive & anti corrosion additives
3. Pump
    a. 12V DC centrifugal pump
    b. Flow rate: 2-4 L/min
    c. Maintains continuous coolant circulation
4. Radiator (120 mm)
    a. 120 mm length fan-compatible radiator
    b. Aluminum fins for high surface area
    c. Air removes heat from coolant



Working Principle



- Step-by-Step Working:
- CPU generates heat (Q)
- Heat conducted to copper water block
- Coolant absorbs heat (Convection)
- Pump circulates hot coolant to radiator
- Radiator + Fan remove heat to air (Forced Convection)
- Cooled liquid returns to CPU

- 🌡️ Heat Transfer Modes Used:
- Conduction → CPU → Water block
- Convection (Internal) → Coolant flow
- Convection (External Forced) → Radiator fan to air

- 📈 Why Liquid Cooling is Better?
- ✅ Higher heat capacity than air
- ✅ More stable CPU temperature
- ✅ Suitable for high-performance & overclocked systems
- ✅ Reduced thermal throttling

Design Parameters & Component Selection

⚙️ Pump Selection:

- Type: Centrifugal DC pump
- Voltage: 12V
- Flow Rate: 2-4 L/min
- Head Pressure: ~2 meters

💧 Coolant:

- Distilled Water + Ethylene Glycol (best thermal performance)
- Reason:
- ✅ Good thermal conductivity
- ✅ Anti-corrosion
- ✅ Safe for closed-loop system

💨 Radiator Design:

- Size: 120 mm length
- Material: Aluminum fins + copper tubes
- Fan speed: 1200-1800 RPM
- Heat dissipation capacity: ~150-250 W

🎯 Expected Outcome:

- CPU temperature reduction by 15-25°C
- Stable operation under heavy load
- Increased CPU lifespan
- Improved performance efficiency

Competitive Landscape

Technology	Performance Level	Cost Level	System Complexity
Air Cooling	Moderate	Low	Simple
Heat Pipe Cooling	High	Medium	Moderate
All-In-One Liquid Cooling	Very High	High	Complex
Custom Liquid Cooling	Excellent	Very High	Very Complex
Proposed Affordable Liquid Cooling System	Very High (Liquid-Level Performance)	Low to Medium	Moderate & Reliable

Liquid Cooling Market Analysis

TAM – Total Addressable Market

Global CPU Thermal Management Market (Air + Liquid + Data Center Cooling).

Estimated Value: $3.5 Billion per year

SAM – Serviceable Available Market

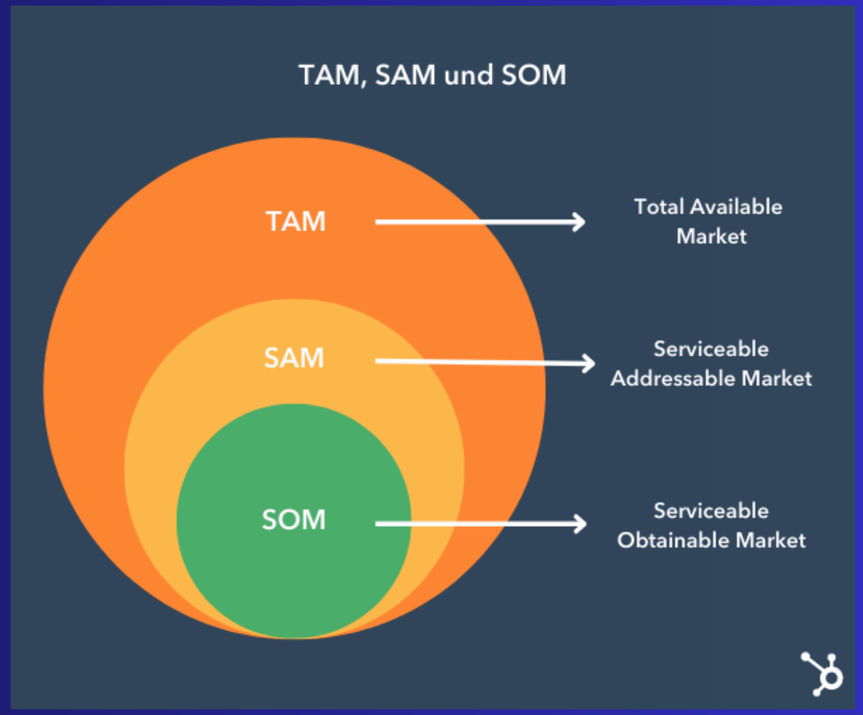
Consumer and Workstation Liquid Cooling Segment (~30% of total market).

Estimated Value: $1.05 Billion per year

SOM – Serviceable Obtainable Market

Targeting 2% penetration of liquid cooling market.

Estimated Value: $21 Million per year



Conclusion

- CPU power density is increasing rapidly (200W-1000W+ range).
- Air cooling is approaching physical heat flux limits (~100-200 W/cm²).
- Advanced cooling systems exist but are expensive and complex.
- Our liquid cooling design improves:
  - Heat transfer area
  - Flow distribution
  - Hotspot management
- Achieves strong thermal efficiency-to-cost ratio without oversizing components

References

Development of Liquid Cooling for PCs

Component Analysis for Factory Sealed Systems

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Development of an Internal Liquid Cooling System for CPU using CFD

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